

NIORO DU SAHEL, Mali

WMO: 612300

Lat: 15.239N

Lon: 9.577W

Elev: 237

StdP: 98.51

Time Zone: 3.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	14.0	-12.7	1.3	26.2	-9.8	1.7	27.0	7.6	25.9	6.8	25.9	0.7	80	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.0	43.4	21.2	42.5	21.4	41.5	21.5	27.5	33.8	26.9	33.0	26.5	32.6	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.2	30.3	25.5	21.3	29.8	25.0	20.7	29.6	89.0	33.9	86.1	32.9	84.3	32.6	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.7	5.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			3.9	29.4	1.2	1.0	3.1	30.1	2.4	30.7	1.7	31.2	0.9	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	22.8	25.8	29.7	33.1	35.8	34.9	30.9	28.9	29.6	31.3	28.1	24.5
	DBStd	4.47	2.48	3.15	3.00	2.37	1.92	2.44	2.13	1.73	1.86	1.51	2.23	2.94
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	2	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7164	398	442	612	693	798	747	647	584	589	661	543	449
	CDD18.3	4125	142	209	354	443	540	497	388	326	339	403	293	191
	CDH23.3	59530	1609	2611	5173	7283	9316	8230	5404	3899	4233	5703	3846	2222
	CDH26.7	37107	715	1448	3282	5052	6864	5856	3084	1802	2158	3458	2268	1119
(14) Wind	WSAvg	2.5	2.1	2.6	2.6	2.6	2.7	3.6	3.1	2.6	2.2	2.1	1.8	2.1
(15) Precipitation	PrecAvg	346	1	1	1	2	6	24	83	140	75	14	1	1
	PrecMax	517	3	10	10	12	22	54	162	227	146	63	8	8
	PrecMin	195	0	0	0	0	0	4	43	49	30	1	0	0
	PrecStd	73	1	2	2	3	6	14	28	38	27	13	2	1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.4	39.9	41.9	43.8	44.5	44.3	40.0	37.0	38.8	40.2	38.9	37.0
		MCWB	16.7	19.6	19.3	20.9	20.6	20.5	24.0	25.7	24.0	20.7	18.2	17.5
	2%	DB	33.6	37.7	40.5	42.5	43.5	43.1	38.2	35.3	36.7	39.3	38.0	35.5
		MCWB	16.0	18.7	19.4	20.4	21.3	21.4	24.7	25.8	25.6	21.1	18.1	17.4
	5%	DB	32.3	36.2	39.2	41.5	42.7	41.9	37.0	34.0	35.4	38.4	37.0	34.3
		MCWB	15.3	18.5	18.9	20.5	21.8	21.9	24.9	25.9	25.9	21.4	18.1	17.0
	10%	DB	30.8	34.5	37.9	40.6	41.8	40.5	35.8	33.0	34.3	37.4	35.9	32.6
		MCWB	14.6	17.3	18.5	20.2	22.0	22.6	25.0	25.8	25.9	22.0	18.2	16.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	25.0	25.1	25.8	29.2	26.6	28.3	28.2	27.8	27.4	24.1	21.1
		MCDB	31.5	35.1	36.5	40.3	39.9	36.1	33.8	32.6	33.9	33.9	32.2	34.1
	2%	WB	18.7	22.7	23.2	24.4	26.6	25.9	26.7	27.3	27.2	26.2	22.2	19.4
		MCDB	29.3	33.3	35.6	38.5	38.2	35.8	33.4	32.1	33.0	32.7	30.0	33.1
	5%	WB	17.1	20.4	21.5	23.2	25.2	25.3	26.2	26.8	26.8	25.6	20.3	18.0
		MCDB	28.8	31.9	34.4	38.1	36.7	35.2	33.0	31.6	32.4	32.4	33.2	31.4
	10%	WB	15.6	18.5	19.9	22.0	24.4	24.7	25.7	26.3	26.4	25.1	19.1	16.8
		MCDB	28.4	32.4	33.6	37.1	36.2	34.8	32.5	31.1	32.0	32.1	33.0	30.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	15.8	15.5	14.8	12.0	10.3	8.8	7.9	9.3	13.0	16.0	15.9
		MCDBR	17.6	17.1	16.2	15.2	13.1	11.6	9.8	9.0	10.7	14.4	16.8	17.2
	5% WB	MCWBR	7.4	7.1	6.5	6.1	4.9	3.7	2.6	2.8	3.3	4.3	5.6	6.7
		MCWBR	16.5	15.7	13.1	14.1	12.1	10.2	8.9	8.0	9.4	11.2	15.2	15.2
(40) Clear-Sky Solar Irradiance	taub		0.505	0.534	0.635	0.647	0.691	0.731	0.670	0.595	0.579	0.574	0.490	0.474
		taud	1.864	1.795	1.595	1.620	1.554	1.542	1.685	1.858	1.875	1.841	1.952	1.964
	Ebn at Noon		781	780	716	709	667	634	676	737	749	738	790	797
		Edh at Noon	196	219	274	268	282	282	245	208	204	206	178	173
(44) All-Sky Solar Radiation	RadAvg		5.25	5.88	6.57	6.92	6.65	6.33	5.96	5.84	6.01	5.87	5.40	5.07
	RadStd		0.32	0.29	0.30	0.29	0.30	0.17	0.24	0.22	0.25	0.14	0.23	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page